



United International University
Dept. of Computer Science & Engineering
Object Oriented Programming Laboratory – CSE 1116
Home Assignment 5

[Q1] Create a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator should have two input fields where users can enter numerical values. Additionally, it should display the result of the operation in a non-editable field below the input boxes. The calculator should include buttons for each operation (+, -, *, /) and a clear button to reset both input fields and the result. **[5]**

Your task is to design a graphical user interface (GUI) using Java Swing that fulfills the following requirements:

- The calculator must have two input fields (a and b) to accept numerical values.
- The calculator should display the result after each operation in a non-editable result field.
- The user must be able to perform addition, subtraction, multiplication, and division.
- The calculator should handle exceptions such as invalid inputs (non-numeric values) and prevent division by zero and other possible exceptions.
- Provide clear error messages to the user when invalid operations are attempted using dialog box.

The screenshot shows a Java Swing window titled "Simple Calculator". The window has a light gray background and a standard title bar with minimize, maximize, and close buttons. Inside the window, there are three text input fields and a row of five buttons. The first input field is labeled "DISPLAY:" and is positioned at the top. Below it are two more input fields, labeled "A:" and "B:" respectively. At the bottom of the window, there are five buttons: a plus sign (+), a minus sign (-), an asterisk (*), a forward slash (/), and a "C" button for clearing. All input fields and buttons are outlined in black.